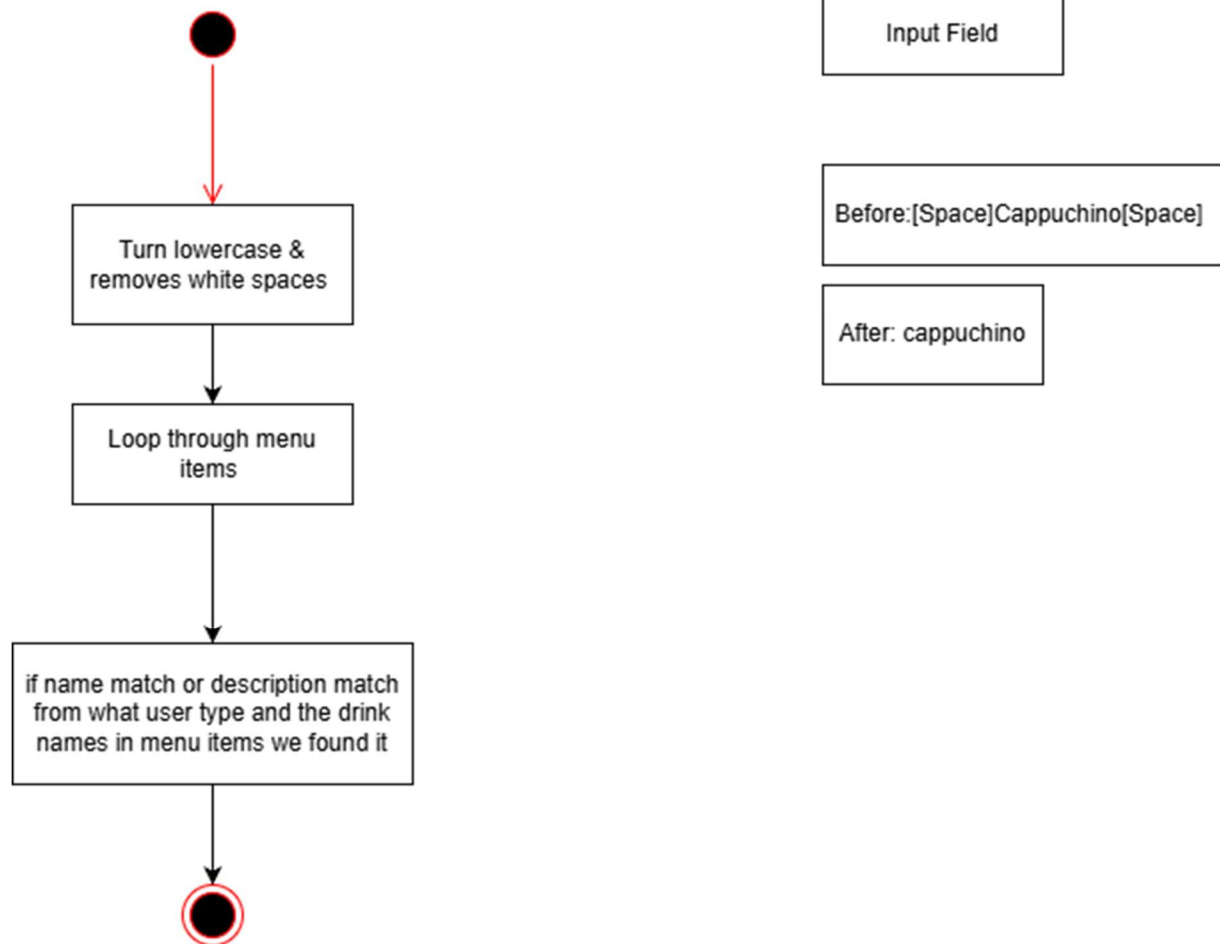


Algorithms and Data Structures

Explain the coding process

Describe the algorithms that implement mathematical models from your previous reports.

Algorithm activity diagrams



Complexity: $O(n * m)$

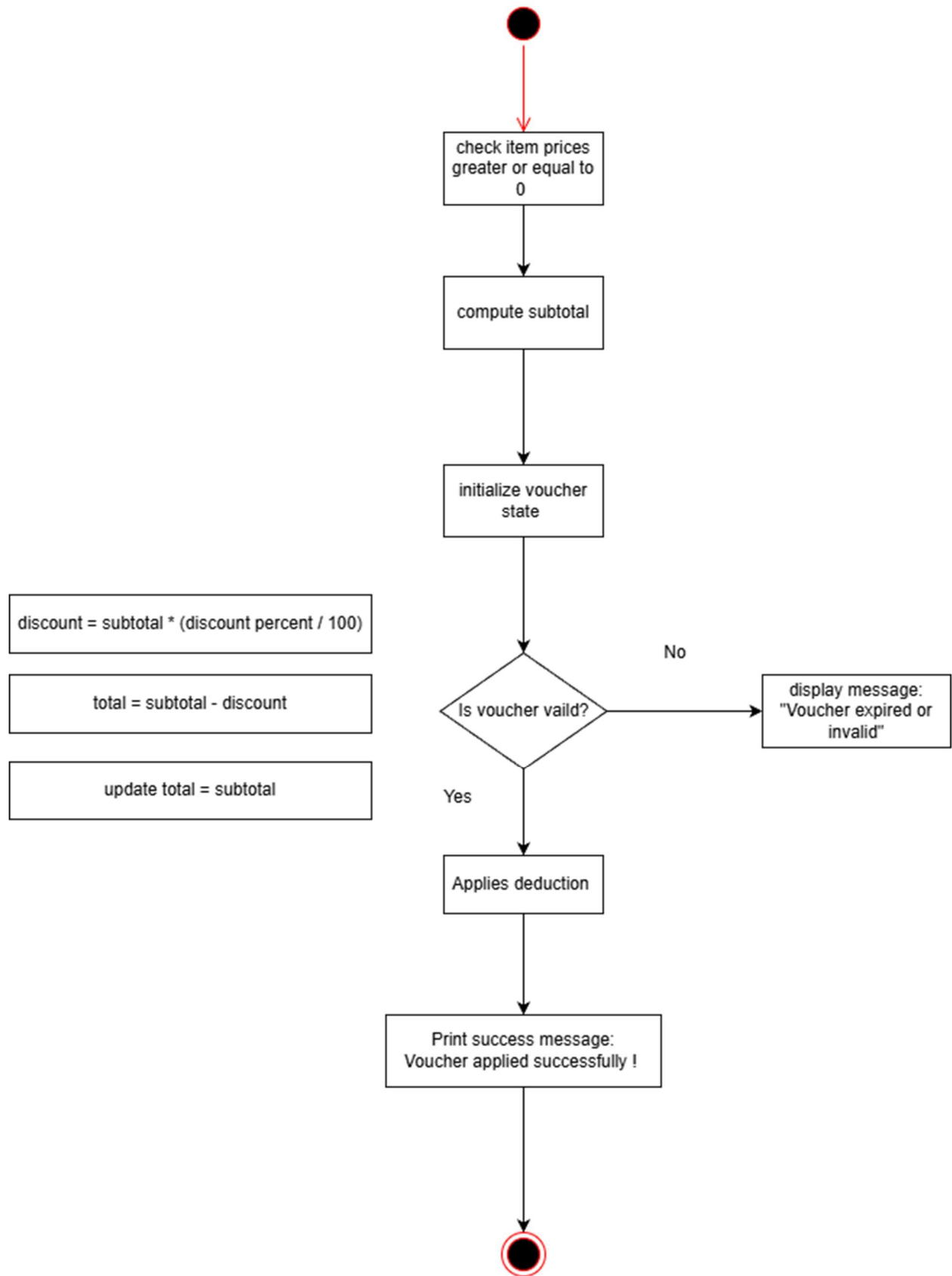
n is where the total number of items listed in the menu

For example: we have 11 drinks list in the menu, n will be 11

m : represent the total characters of the user type in the input field

for example: cappuchino = 10

$$O(n*m) = 11 * 10 = 110$$



Does your system use any complex data structures, such as arrays, linked list, hash tables, or trees?

Arrays

In my coding process I use arrays including in menu Items, cart and to filter item.

I rely on time complexity and space complexity. In this case the array operations I use perform $O(n)$. One reason why I choose arrays to do implements this is because arrays provide fast indexing to maintain and reusable structure for storing data efficiently. So, I would need to initialize the arrays, use the algorithm to process storing from menu Items.

I believe using a linked list would be more robust and useful than Arrays in my case, because the arrays are in $O(n)$ run time, if I implement a doubly linked list, I can use insertion at front and deletion at end to make it $O(1)$ run time. A better data structure that not only works but is reliable for this specific type of problem.

Hash Tables $O(1)$

I chose Hash table because it gives faster look up, I plan to use it in back end, for functionality like getting

Linked list $O(1) \sim$ head and tail

I chose linked list because it is going to be used in insertion and order deletion for individual order transactions

Fibonacci heaps $O(1)$

I chose Fibonacci heaps because it prioritizes and optimizes scheduling, and is efficient in storing data

Queue $O(1)$

I chose Queue for my system because I treat it as how often the user revisits the system and how long does it take for the user to complete the service.

Discussion: for the metric that I use to measure the effectiveness of complex data structures, I choose space complexity and average time complexity to see how well it works in real time.

For performance check and how to measure it technically, what I would do is find the execution time – how long does it takes for Queue, Fibonacci heaps, linked list, hash tables and arrays completes it task by taking the time after executing complex data structure minus time before we begin to execute it.

